

Harrison Davis

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Detail-oriented Mechanical Engineering student who thrives on taking ideas from concept through implementation. Pursuing a career that allows me to apply and grow my knowledge and experience on projects that solve problems and improve outcomes.

Education

Embry-Riddle Aeronautical University, Daytona Beach, FL
Bachelor of Science, Mechanical Engineering - Robotics-Focus (Sophomore)

Expected Graduation: May 2028
Current GPA: 3.74

Rowan-Cabarrus Community College, Concord, NC
Associates of Science

Dec, 2023
GPA: 3.93

Technical Skills

- **Engineering and Programming Software:** CATIA v5, Lua, MATLAB, Python
- **Office Software:** Excel, PowerPoint, Word, Illustrator, Photoshop, Clip Studio Paint
- **Hardware:** Additive Manufacturing, CNC Milling, Lathe Operation, Prototyping, Soldering
- **General:** Collaboration, Design, Building, Programming, Testing, Trouble-Shooting, Team Building

College Engineering Project Experience

- Led a team of five to design and build a Halloween animatronic in EGR-101.
- Led a team of three to diagnose, brainstorm, and implement a fix for an ERPL engine test stand.

Self-Directed Project Experience

Software Projects

- **3D Rendering Engine:** Developed a 3D engine to learn and apply the math behind computer graphics (dot products, matrix multiplication, cross products)
- **Neural Network:** Built a small layered neural network, using an “evolutionary” genetic algorithm.
- Competed in several game-jams, working under time limits.

Physical Projects

- **Unmanned Aerial and Ground Vehicle (UAV and UGV) Building:** Designed and built multiple fixed-wing UAVs and numerous UGVs of my own design over the past 12 years.
- **Computer Hardware:** Built 4 desktop PCs. Restored 6 laptops and computers with hardware damage to functional condition. Repaired several smartphones, an iPod, miscellaneous appliances and other devices.
- Made countless Arduino powered, Raspberry Pi powered, Feather, Ti-84+, etc... projects for personal use and learning.

Work Experience

Table 11 Restaurant, Concord, NC

June 2021 – June 2024

- Maintained kitchen machinery/appliances, prepared/cooked food, adapted to constantly changing priorities.

Protodyne Machine Shop, Harrisburg, NC

August 2017 – February 2020

- **Machine Work:** Maintained and operated shop CNC mills and lathes.
- **Manual Work:** Manually sanded parts using band saw, cut metals, ensured finished products met acceptable deviations.

Club Involvement

- **Experimental Rocket Propulsion Labs:** Structures Team Member, Chassis Redesign Leader
- **Barefoot Flying Club:** Member
- **Game Development:** Member

Hobbies

- **Classic Hardware & Software:** I enjoy working on older machines with limited capabilities, Drawing/Sketching, Kayaking